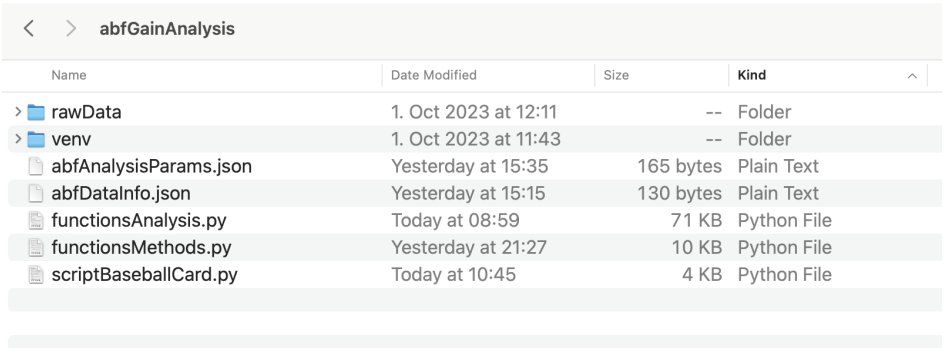
Instructions for Using .abf Data Analysis Python Scripts

The working directory of the project is *abfGainAnalysis/* - this can be renamed as desired - and initially should hold the following files; these file names **cannot** be altered as they are explicitly imported.

- abfAnalysisParams.json
- abfDataInfo.json
- functionsMethods.py
- functionsAnalysis.py
- scriptBaseballCard.py

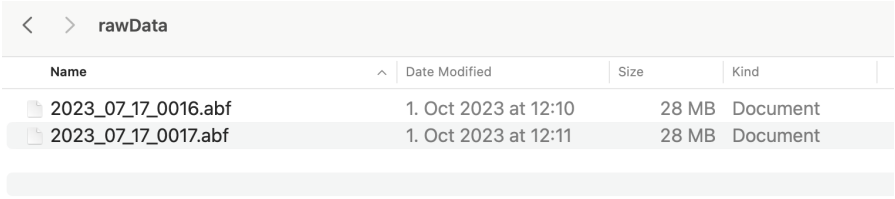
All .abf data files must be saved in a folder, in the working directory, named *rawData/*. This folder must be created and named manually by the user. It is the only place the python script will search for data files. If using an IDE a *venv/* folder is automatically created as soon as you start a new project. If using PyCharm (or a similar program) the *abfGainAnalysis/* project initially is as shown below.



The user must manually install the requisite Python modules: scipy, matplotlib, natsort, pyabf, seaborn. This can be accomplished by using the IDE's local terminal and typing the following line:

```
PIP INSTALL SCIPY MATPLOTLIB NATSORT PYABF SEABORN
```

For exemplary analysis two .abf data files were analyzed and baseball cards (output figures) were created and saved; specifically the *rawData/* folder holds two data files as shown below.



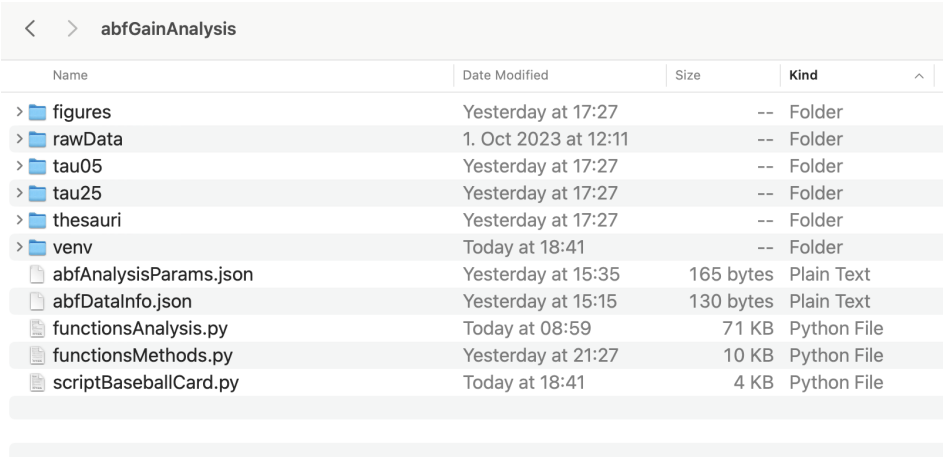
The user must now open *abfDataInfo.json* and manually enter the data file names to be analyzed into "dataNames". One can copy and paste each file name and replace .abf with a forward slash /. Each file is enclosed in double quotes and separated by a comma.

The user must manually set the "pathName" in the *abfDataInfo.json* file. This is the location of the working directory. Simply right-click on any of the .py files and viewing settings (e.g. 'get info' for macOS, 'properties' for Windows and Linux) will show the working directory path.

For this exemplary analysis the *abfDataInfo.json* file appears as shown below.



Nothing else is required at this point. The data the user has saved in *rawData/* and indicated the names of and path for in the *abfDataInfo.json* file will be analyzed. For each data file all folders are automatically created in the working directory. A new folder called *figures/* is created and each baseball card is saved as .png (portable networks graphic) and .svg (scalable vector graphics). For the two exemplery data files after analysis the working directory now appears as shown below.



The two exemplary baseball cards are displayed on the overleaf. Descriptions about variables names used in the code, information contained in outputs, and other notes can be found in the extensive comments included in each of the three .py files. JSON files by convention do not contain comments.

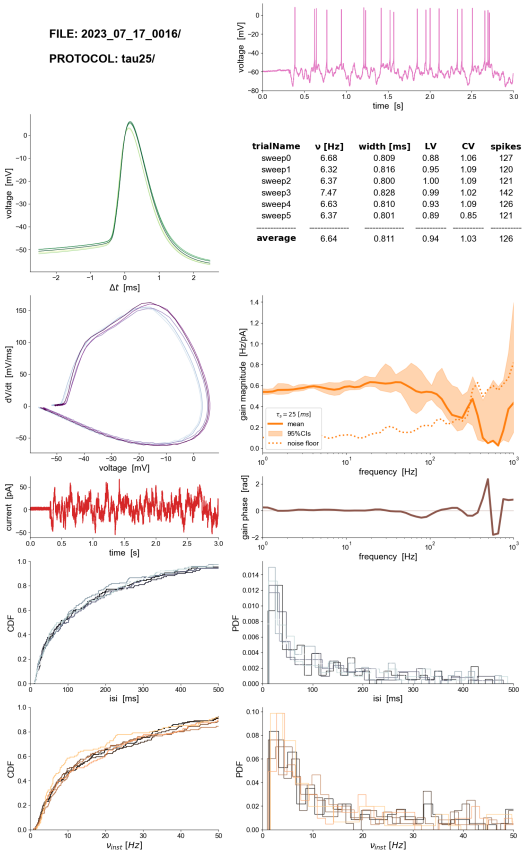
If different analysis parameters are desired, they may be altered by the user by opening *abfAnalysisParams.json* and making changes. It does not matter whether numbers are entered as ints or floats; they are converted to the correct type when imported. They are described at the top of *sciptBaseballCards.py*.

Changes to *functionsMethods.py* and *functionsAnalysis.py* are strongly discouraged.

This code only works properly on .abf files which have been run using an Ornstein-Uhlenbeck input protocol labelled using the format OU_25ms. If the protocol name is non-compliant the program exits and the error message "incorrect protocol name" is displayed.

FILE: 2023_07_17_0016/

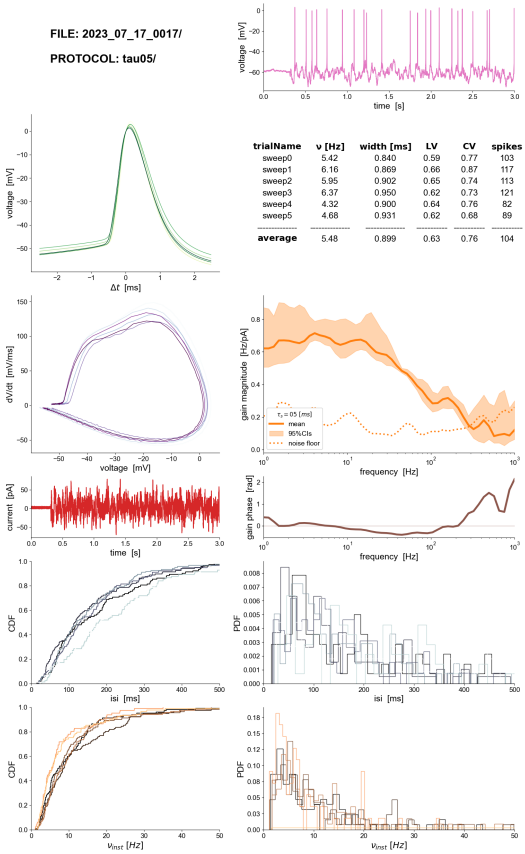
PROTOCOL: tau25/



trialName	ν [Hz]	width [ms]	LV	CV	spikes
sweep0	6.68	0.809	0.88	1.06	127
sweep1	6.32	0.816	0.95	1.09	120
sweep2	6.37	0.800	1.00	1.09	121
sweep3	7.47	0.828	0.99	1.02	142
sweep4	6.63	0.810	0.93	1.09	126
sweep5	6.37	0.801	0.89	0.85	121
average	6.64	0.811	0.94	1.03	126

FILE: 2023_07_17_0017/

PROTOCOL: tau05/



trialName	ν [Hz]	width [ms]	LV	CV	spikes
sweep0	5.42	0.840	0.59	0.77	103
sweep1	6.16	0.869	0.66	0.87	117
sweep2	5.95	0.902	0.65	0.74	113
sweep3	6.37	0.950	0.62	0.73	121
sweep4	4.32	0.900	0.64	0.76	82
sweep5	4.68	0.931	0.62	0.68	89
average	5.48	0.899	0.63	0.76	104